CHAPTER 1 FLOW BASICS

1.1 Little's Law

Little's Law states that work in process equals throughput multiplied by cycle time. Written as a formula, WIP = TH x CT. It holds for any stable production line regardless of the arrival or service distributions.

Managers use Little's Law to estimate cycle time when WIP and throughput are known. For example, a line with 120 jobs in process and a throughput of 40 jobs per day has an average cycle time of three days.

1.2 Variability

Variability degrades the performance of a production system. Higher variability in arrivals or processing times increases queue time, which in turn increases cycle time. The coefficient of variation is the standard deviation divided by the mean.

Buffers against variability take three forms: inventory, capacity, and time.